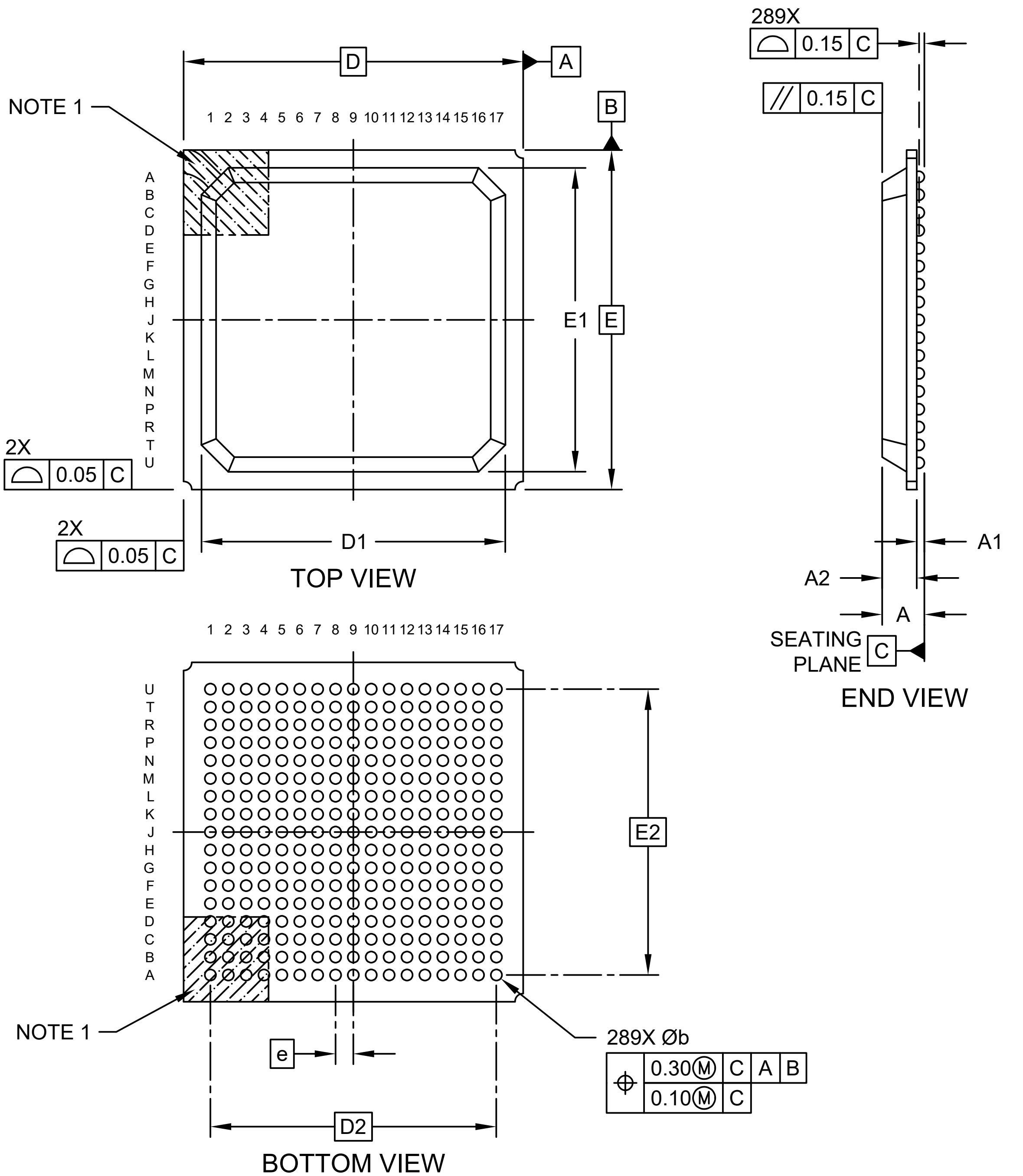
289-Ball Plastic Ball Grid Array Package (ACA) - 19x19x2.36 mm Body [PBGA]
Micrel Legacy Package PBGA19x19-289L

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



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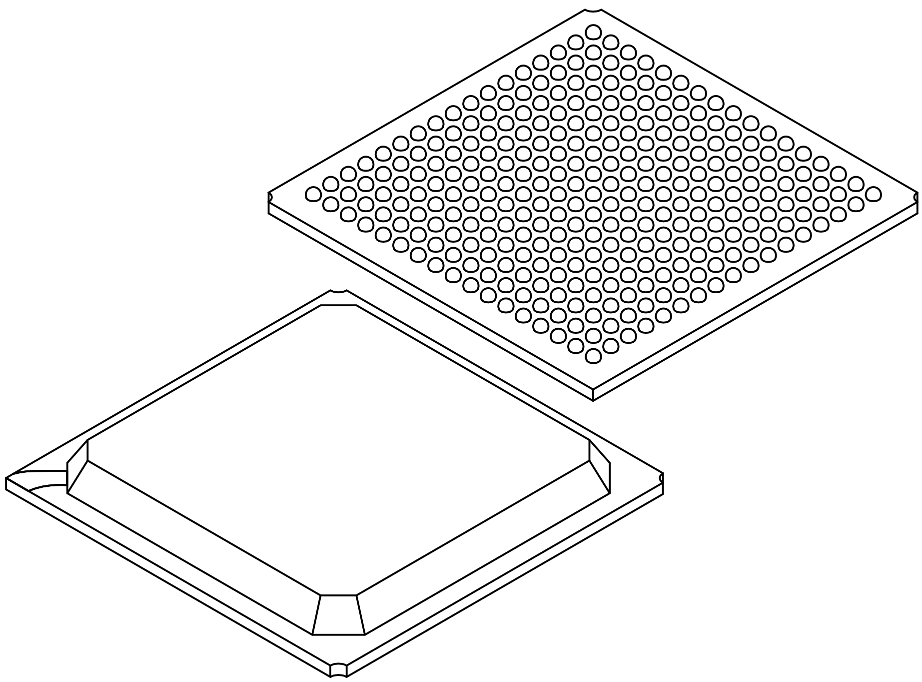

	Units	MILLIMETERS		
	Dimension Limits	MIN	NOM	MAX
Number of Terminals	N	289		
Pitch	e	1.00 BSC		
Overall Height	A	–	2.21	2.36
Ball Height	A1	0.43	0.48	0.53
Package Thickness	A2	1.63	–	1.83
Overall Length	D	19.00 BSC		
Mold Cap Length	D1	16.90	17.00	17.10
Ball Array Length	D2	16.00 BSC		
Overall Width	E	19.00 BSC		
Mold Cap Width	E1	16.90	17.00	17.10
Ball Array Width	E2	16.00 BSC		
Ball Diameter	b	0.58	0.60	0.62

Notes:

1. Pin 1 visual index feature may vary, but must be located within the hatched area.
2. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.
   REF: Reference Dimension, usually without tolerance, for information purposes only.

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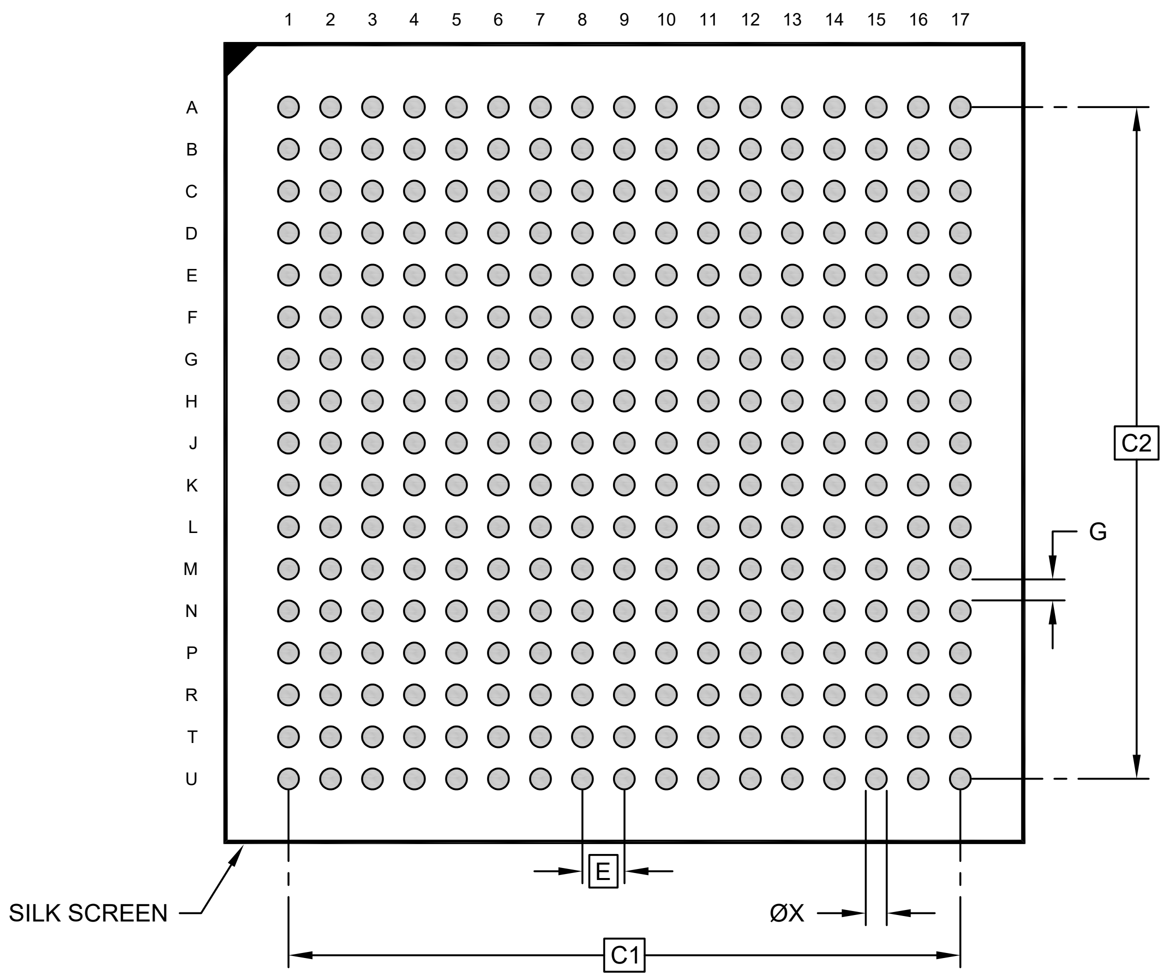

RECOMMENDED LAND PATTERN

	Units	MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E		1.00 BSC	
Contact Pad Spacing	C1		16.00 BSC	
Contact Pad Spacing	C2		16.00 BSC	
Contact Pad Diameter (X289)	X			0.50
Contact Pad to Contact Pad	G	0.50		

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.
2. For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-3087 Rev A